

The Living World

NCERT Line by Line | Detailed Notes for NEET

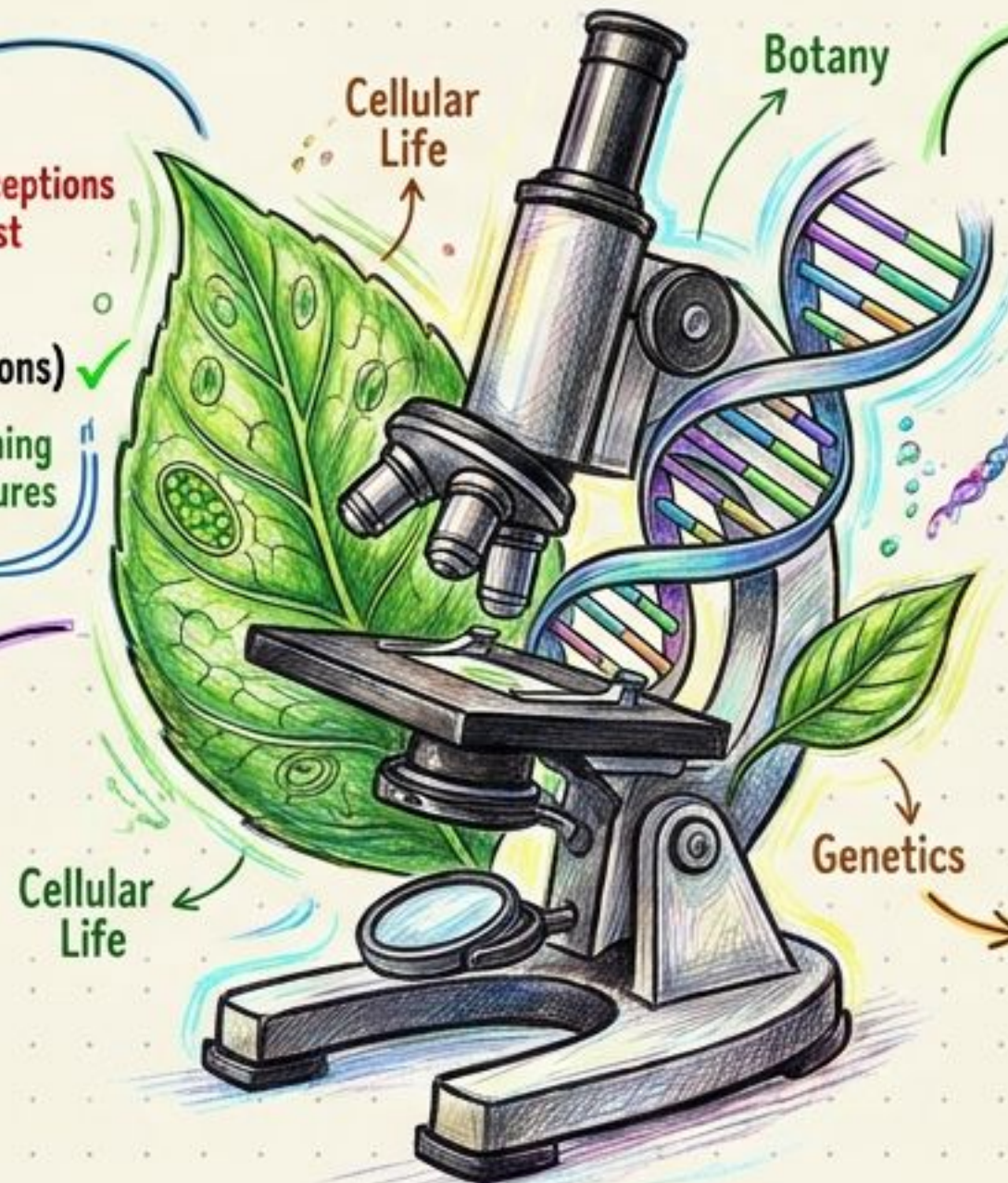
Syllabus Update:
Includes What is Living, Diversity, and Taxonomic Hierarchy (Taxonomical Aids removed from NCERT).

What is Living? (Key Characteristics)

- **Growth** (Not defining property, e.g., sand mounds) ✗ **Exceptions exist**
- **Reproduction** (Not defining property, e.g., mules, sterile worker bees) ✗ **exist**
- **Metabolism** (Defining property, sum of all chemical reactions) ✓
- **Cellular Organization** (Defining property) ✓ → **Defining Features**
- **Consciousness** (Defining property, ability to sense environment)

Taxonomic Hierarchy (Ranks)

- Kingdom (Highest) 👑
- Phylum/Division 🦋
- Class 👑
- Order 🦋
- Family 🐾
- Genus 🦋
- Species (Lowest, basic unit) 🦋



Diversity in the Living World (Biodiversity)

- Number of species known: 1.7-1.8 million
- Need for classification to standardize study
- **Nomenclature** (Naming), **Identification**, **Classification**

Binomial Nomenclature

- Generic name (capitalized) + Specific epithet (lowercase)
- Latin origin, written in Italics or underlined separately
- e.g., *Mangifera indica* (Mango)

Key Points & Mnemonics

- **Keep Pots Clean Or Family Gets Sick** (Mnemonic for Hierarchy)
- **Taxon**: Any rank in hierarchy
- **Systematics**: Study of diversity & evolutionary relationships
- **Taxonomy**: Principles & procedures of classification

What is Living



Characteristic Feature

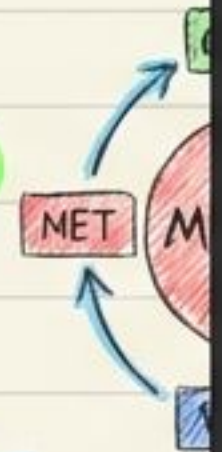
- Definition: Any feature found in a living organism (e.g., walking, dancing, green color).
- Condition: May be present in some living things, but not all.
- Conclusion: Does NOT definitively distinguish living from non-living.



Defining Feature



- Condition 1: 100% present in ALL living organisms.
- Condition 2: 0% present in ANY non-living thing.
- Examples: Metabolism, Cellular Organization, Consciousness.

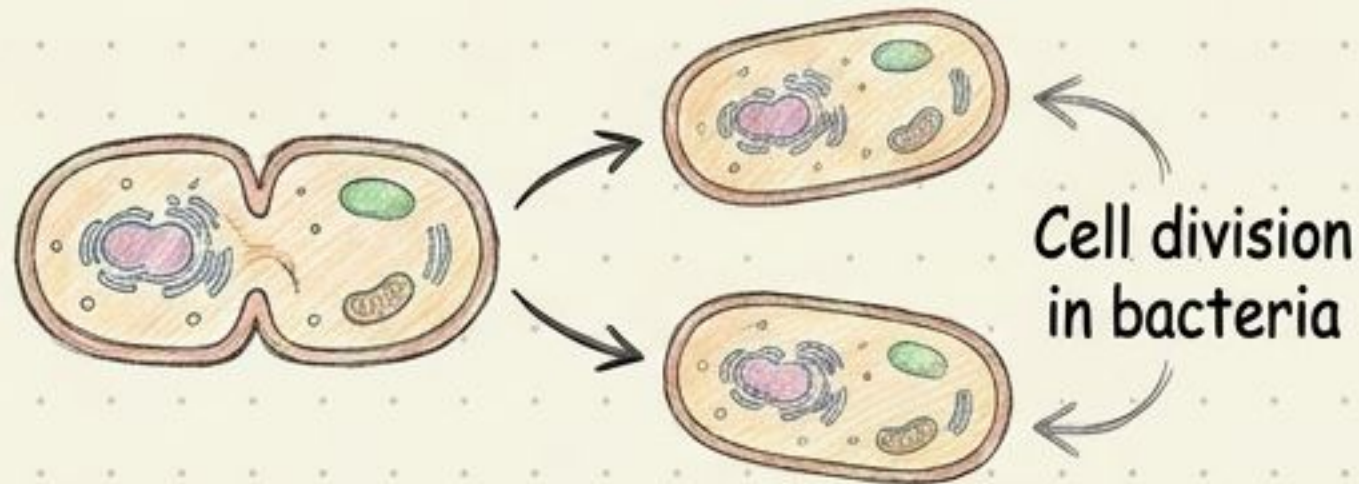


Features without exceptions!
These define life.

Growth

The Twin Characteristics

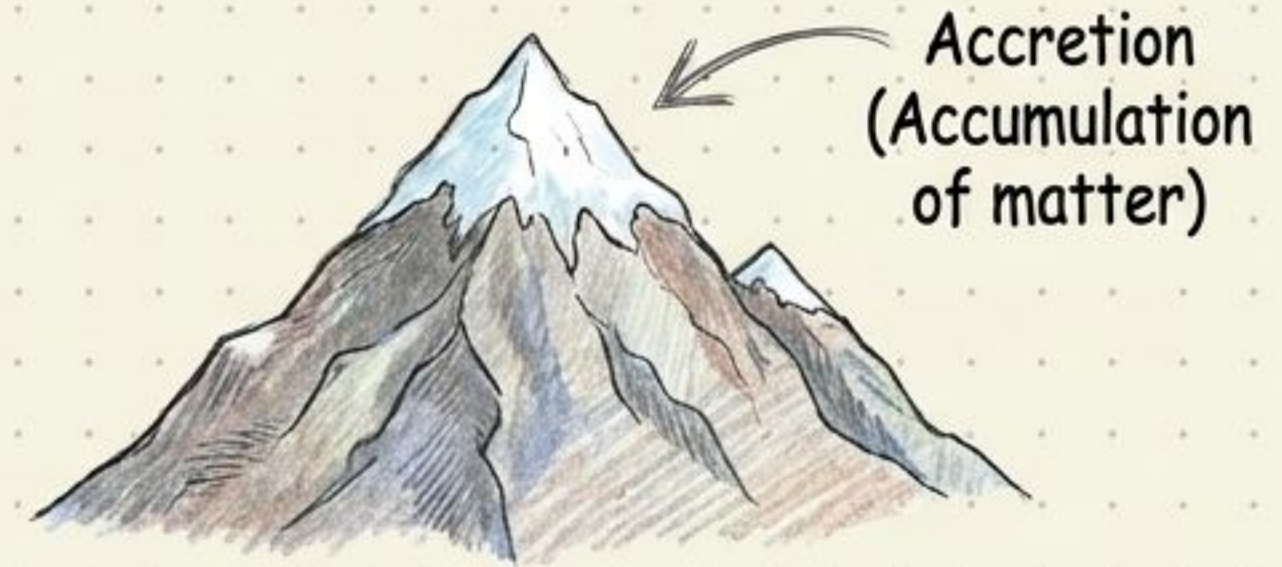
1. Increase in mass
2. Increase in number of individuals



Cellular Relationships:

- Unicellular: Growth, Cell Division, and Reproduction are **Mutually Inclusive (Synonymous)**.
- Multicellular (Higher Organisms): Growth and Reproduction are **Mutually Exclusive**.

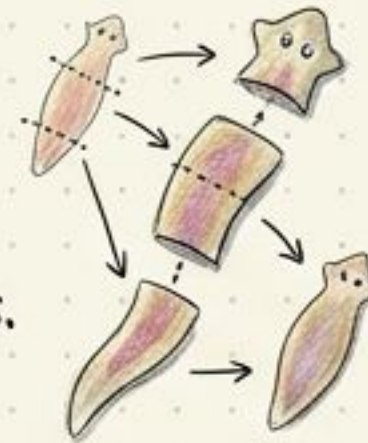
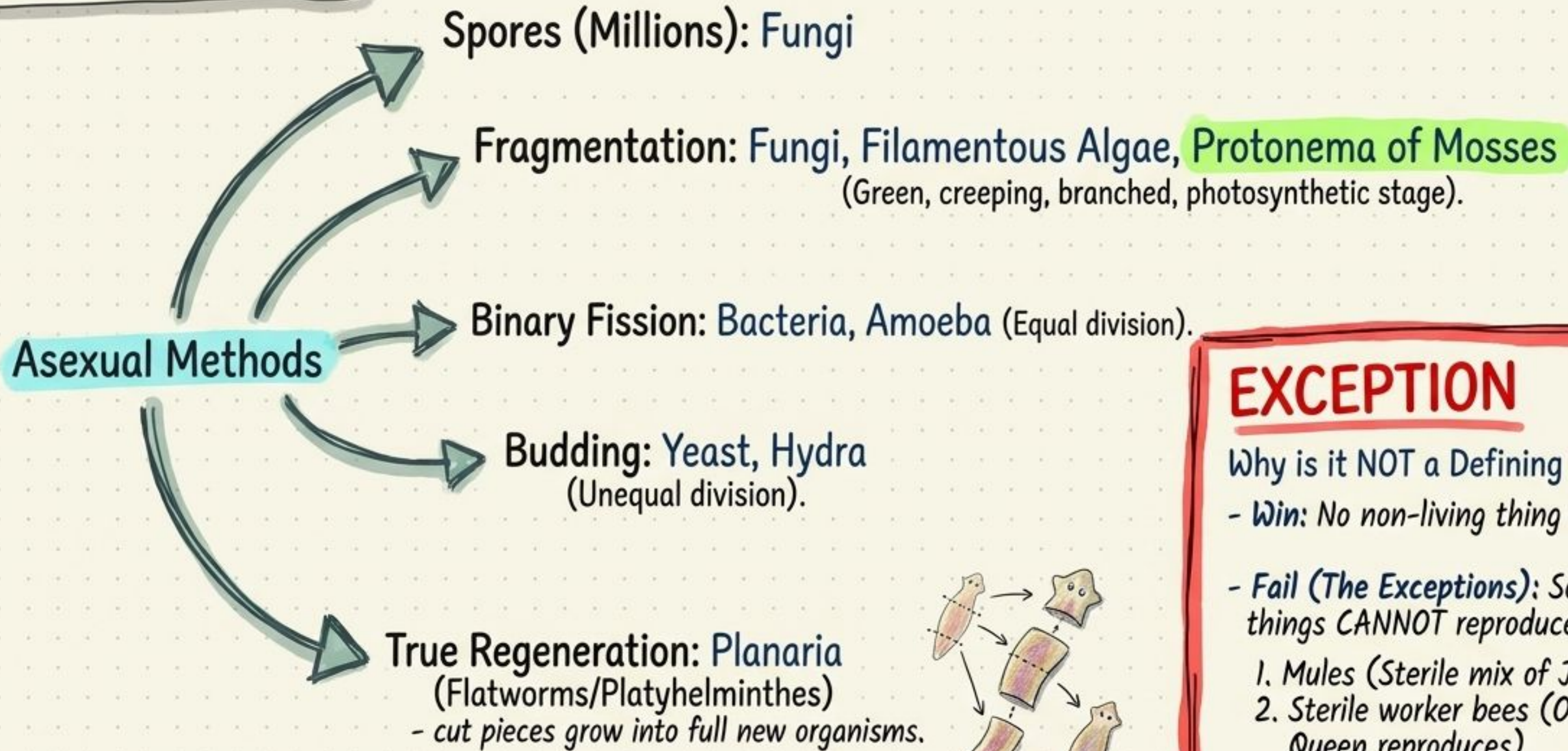
Why is it NOT a Defining Feature?



- Non-living objects (mountains, sand mounds, boulders) also grow by **accumulation of matter on the surface**.

NEET Insight: **Internal Growth** (growth from inside) **CAN** be taken as a **defining feature**, but general 'Growth' cannot.

Reproduction



EXCEPTION

Why is it NOT a Defining Feature?

- **Win:** No non-living thing reproduces.
- **Fail (The Exceptions):** Some living things CANNOT reproduce:
 1. Mules (Sterile mix of Jack & Mare)
 2. Sterile worker bees (Only the Queen reproduces)
 3. Infertile/Sterile human couples.

Metabolism and Cellular Organization

Metabolism (Defining Feature - No Exceptions)

- **Definition:** Sum total of ALL chemical reactions in the body.

Catabolism: Breaking down large molecules (Releases energy / Exothermic / Exergonic).



Anabolism: Building large molecules (Consumes energy / Endothermic / Endergonic).

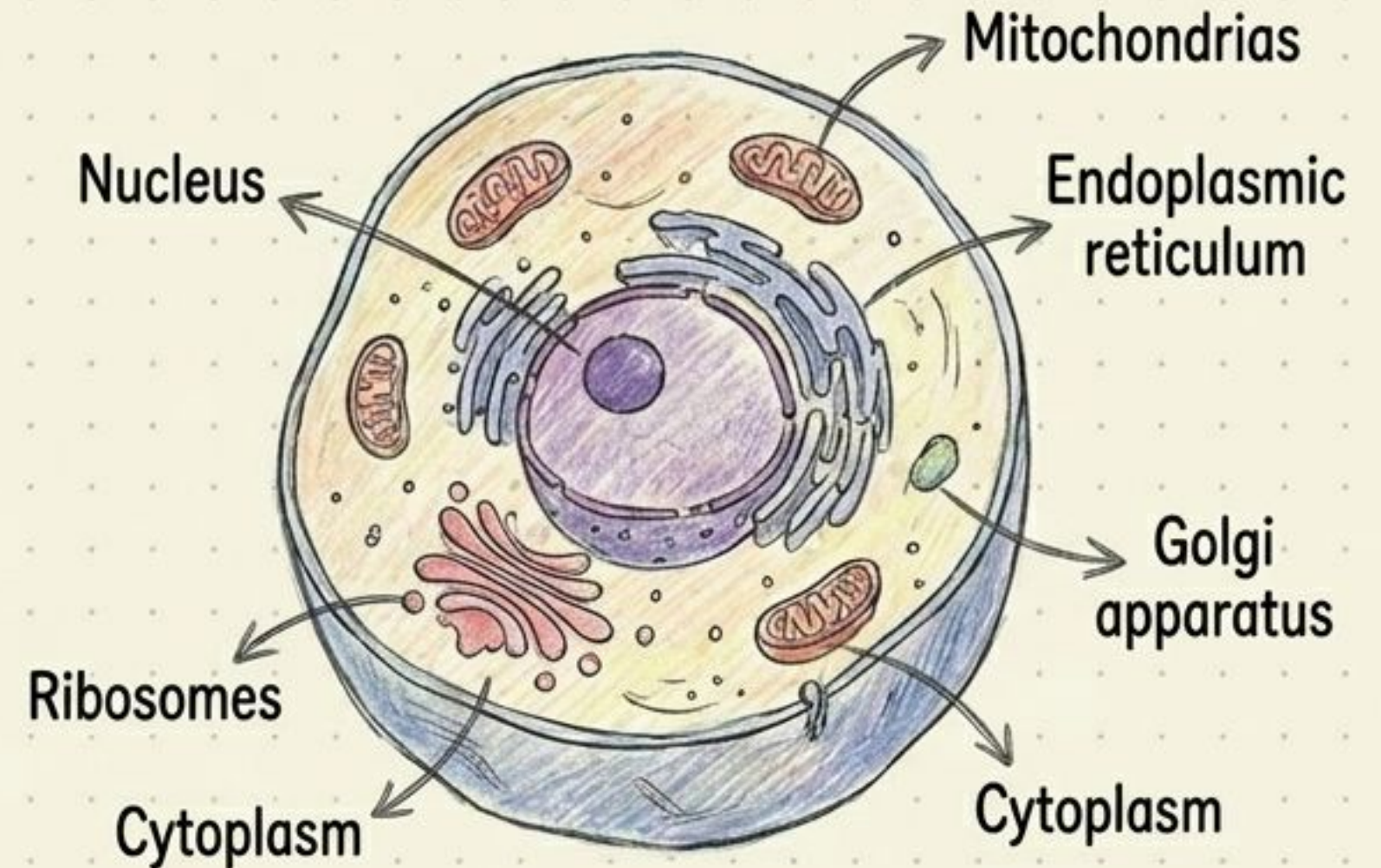


The *In vitro* Exception Rule:

- An isolated metabolic reaction outside the body (in a test tube) = Neither a living nor a non-living thing.
- BUT, it is surely a **Living Reaction**.



Cellular Organization (Defining Feature)



- At least one cell is mandatory for independent life. Without it, survival is impossible.

Consciousness

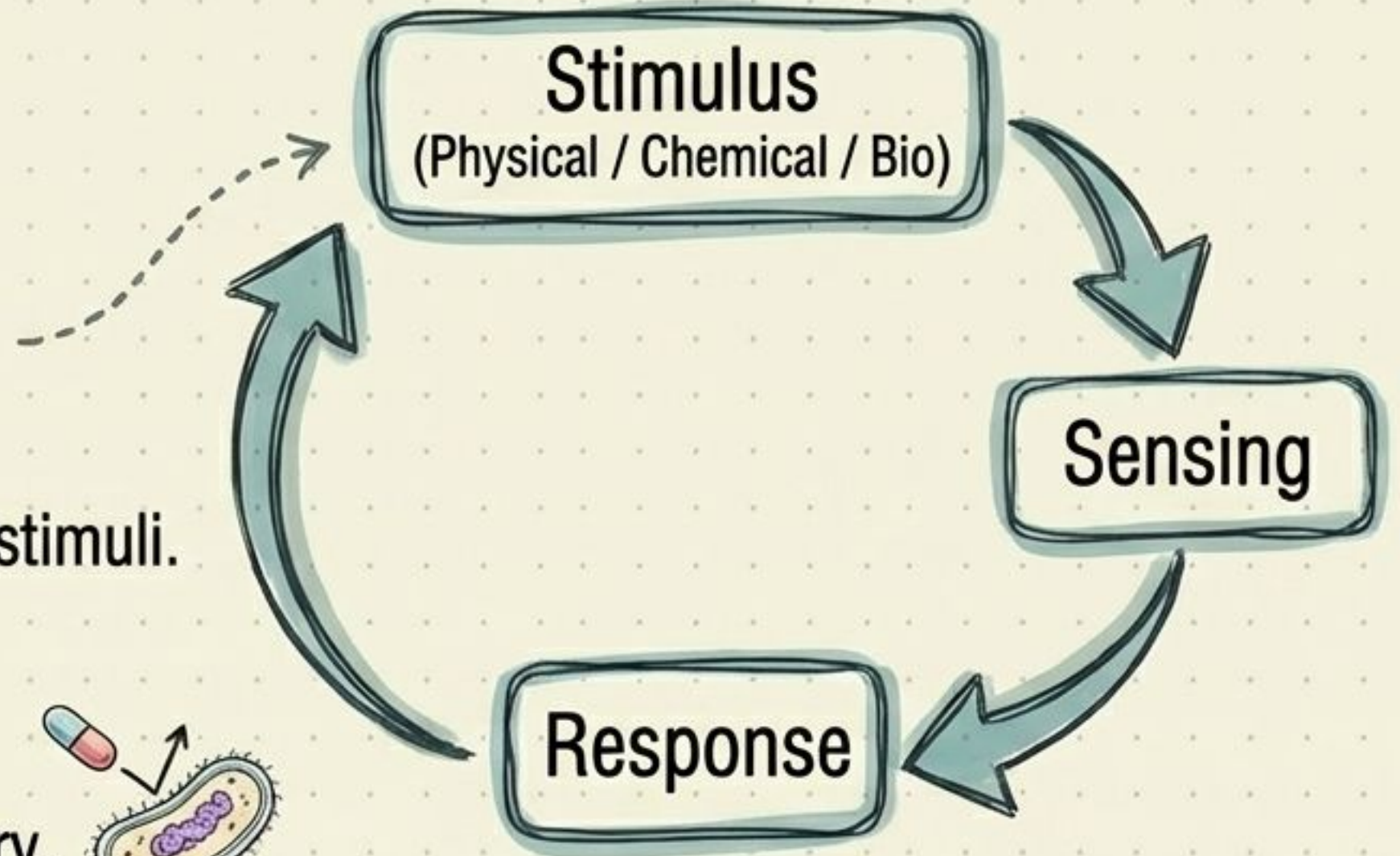
Consciousness (The 'Baap' Feature):

The most complicated and obvious feature of all living organisms (Prokaryotes to Eukaryotes).

Ability to sense the environment and respond to stimuli.

Examples in Nature:

- **Bacteria:** Locks membrane to prevent drug entry.
- **Plants:** Bend towards light, close stomata during water scarcity, Venus Fly Trap capturing insects via osmotic cell changes.
- **Photoperiodism:** Light duration affects reproduction in **Seasonal Breeders** (both plants and animals).



Self-Consciousness:

- Awareness of oneself.
- NOT defining: Only found in Humans.

Biodiversity

Final Definition of Living Organisms:
Self-replicating, evolving, and self-regulating interactive systems capable of responding to external stimuli.



1.7 to 1.8 Million

- Number of species known and described: 1.7 to 1.8 Million (17-18 Lakh).
- > 1.2 Million of these are Animals.
- **Scale Check:** Humans have 8 Billion population, but represent just ONE single species (*Homo sapiens*).

Need for Standardization:

- Vernacular (local) names create confusion globally. We require Universal Scientific Names.



Binomial Nomenclature

Given by: Carolus Linnaeus
(Original name: Carl von Linné).

Naming Codes:

- ICBN (Plants)
- ICZN (Animals)
- ICNCP (Cultivated Plants)
- ICVCN/ICTV (Viruses)
- ICNB (Bacteria)

4 Universal Rules

1. Origin: Latin

(Dead language, meanings won't change over centuries).

2. Two Components:

Generic name (Genus, Capital letter) + Specific epithet (Species, small letter).

Mangifera indica Linn.

3. Formatting:

Hand-written = separately underlined.
Typed = printed in Italics.

4. Author Citation:

Author's abbreviated name at the end in Roman script (e.g., Linn. for Linnaeus).

Rule of Priority:

The first validly published name is universally accepted.

Taxonomy

Taxa / Taxon: A category, rank, or unit of classification. Groups organisms based on similarities/differences.

Conventional Taxonomy (4 Pillars - Strict Sequence)

1. **Characterization:** Studying all features.

2. **Identification:** Recognizing what it is similar to.

3. **Classification:** Placing it in convenient categories.

4. **Nomenclature:** Giving it a scientific name.

Modern Taxonomy (5 Pillars - Mnemonic)

A - Anatomy (Internal structure)

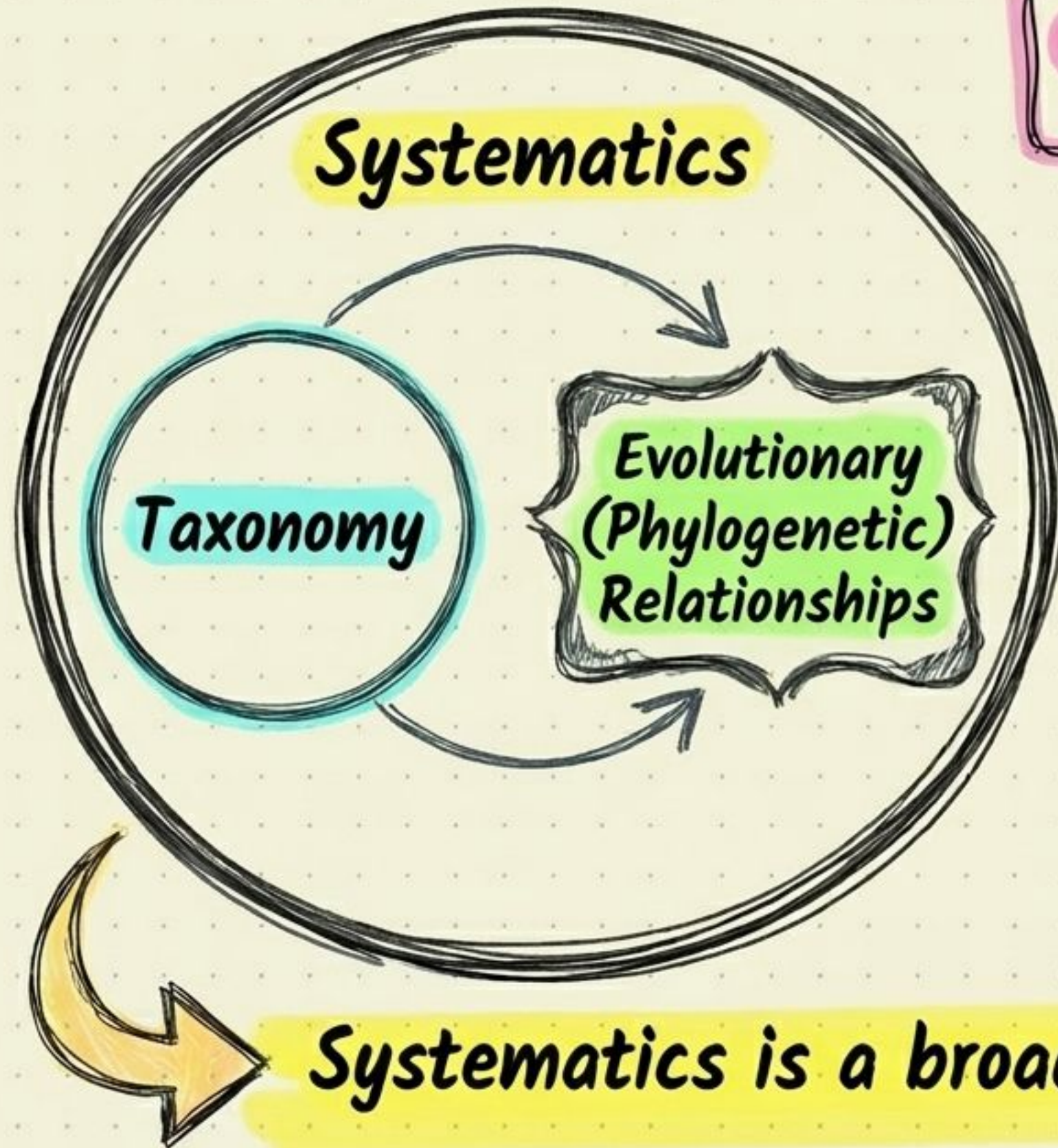
B - Morphology (External structure / Bahar ka)

C - Cell structure

D - Development process

E - Ecological information

Systematics



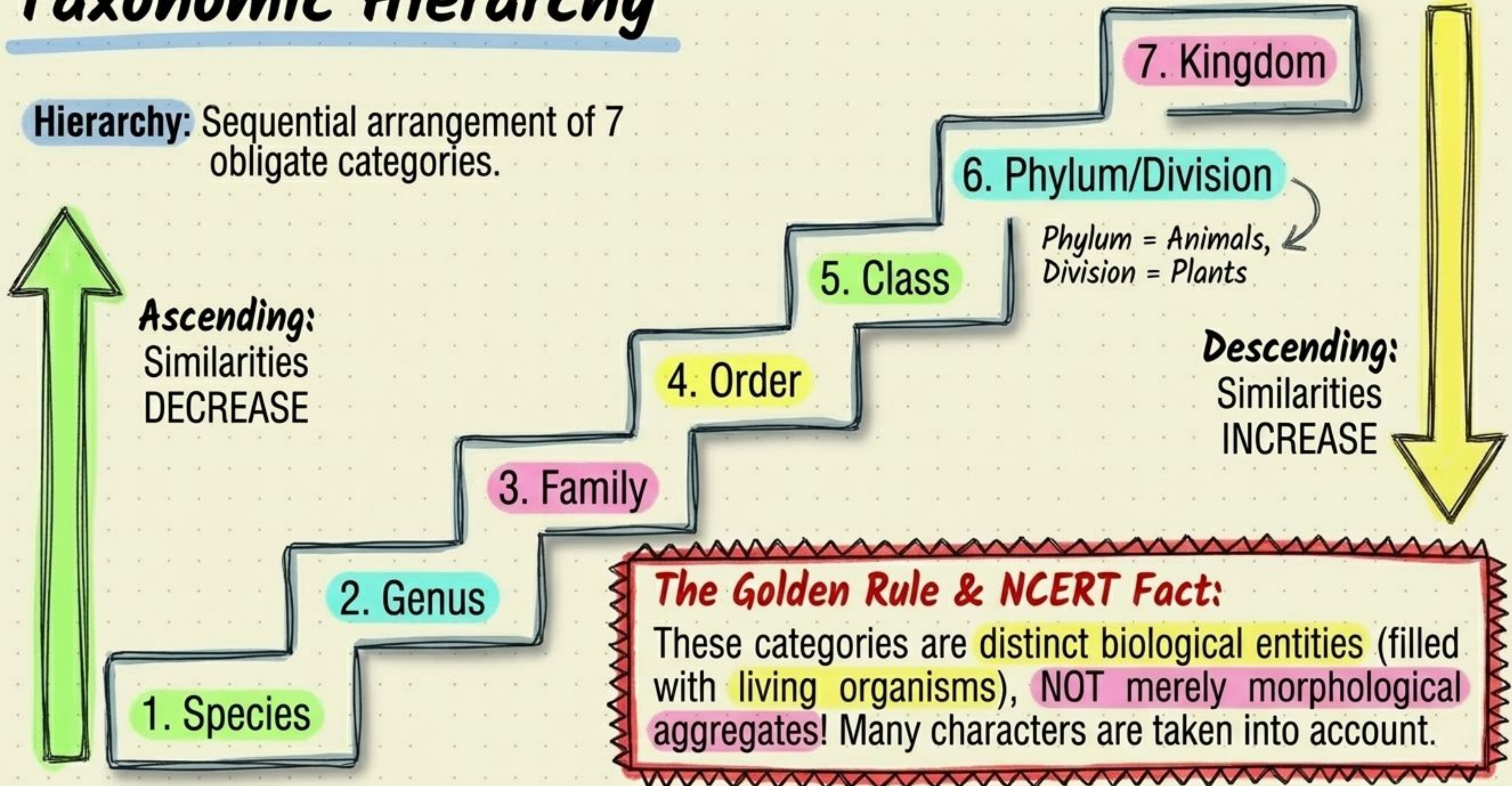
Systematics = Taxonomy + Evolutionary Relationships

- **Key Concept:** Compares DNA/proteins to find common ancestors (Bioinformatics).
- **Origin:** Derived from Latin word *Systema* (Systematic arrangement of organisms).
- **Historical Note:** Linnaeus used *Systema Naturae* as the title of his publication.

Systematics is a broader branch of biology than Taxonomy!

Taxonomic Hierarchy

Hierarchy: Sequential arrangement of 7 obligate categories.



Taxonomic Categories

Species

- Fundamentally similar organisms that interbreed to produce fertile offspring.
- Possesses the maximum number of similarities.

Genus

- Group of related species.
- **Monotypic**: Contains one single species (e.g., *Homo sapiens*).
- **Polytypic**: Contains multiple species (e.g., *Solanum* includes potato, brinjal, makoi; *Panthera* includes lion, tiger, leopard).

Family

- Group of related genera.
- Plant Rule: Classified based on BOTH vegetative (root/stem/leaf) and reproductive (floral) features.



Taxonomic Categories (Suffix Hacks)

Order: Assembly of families based on an aggregate (average) of characters.
(Fewer similarities found here compared to lower taxa).

Cheat Sheet

Plant Suffix Tricks

- Order: Ends in **-ales**
(e.g., Sapindales, Poales, Polymoniales)
- Family: Ends in **-aceae**
(e.g., Anacardiaceae, Poaceae, Solanaceae)

Animal Suffix Tricks

- Family: Ends in **-idae**
(e.g., Hominidae, Muscidae, Felidae)

Key Intersections (NEET Logic):

- *Panthera* + *Felis* = Family Felidae.
- Family Felidae + Family Canidae = Order Carnivora.
- *Solanum* + *Petunia* + *Datura* = Family Solanaceae.
- Family Solanaceae + Convolvulaceae = Order Polymoniales.

Taxonomic Hierarchy (NCERT Data Matrix)

Common Name	Biological Name	Genus	Family	Order	Class	Phylum/ Division
Man	<i>Homo sapiens</i>	<i>Homo</i>	Hominidae	Primata	Mammalia	Chordata
Housefly	<i>Musca domestica</i>	<i>Musca</i>	Muscidae	Diptera	Insecta	Arthropoda
Mango	<i>Mangifera indica</i>	<i>Mangifera</i>	Anacardiaceae	Sapindales	Dicotyledonae	Angiospermae
Wheat	<i>Triticum aestivum</i>	<i>Triticum</i>	Poaceae	Poales	Monocotyledonae	Angiospermae

Must memorize!

The Living World (Wrap-up)

High-Yield Revision Checklist:

- ✓ Defining Features: Metabolism, Cellular Organization, Consciousness (The ultimate marker). Growth and Reproduction are non-defining.
- ✓ Binomial Rules: Latin origin, italicized/underlined, Genus (Capital) + species (small) + Author citation.
- ✓ Hierarchy Flow: Ascending (Species → Kingdom) = Decreasing similarities.
- ✓ NCERT Table: Memorize the Housefly Order (Diptera) and Wheat Family (Poaceae).

→ Base is built. Focus on high-weightage chapters next to crack NEET! 😊